GANITA PRAKASH • CHAPTER 6

Class 7 Mathematics • Part 1

NUMBER PLAY

Every *Figure it Out* solved ⟶ learn ⟶ revise

O E O
Parity patterns

8 1 6
3 5 7
4 9 2
Magic sum 15

1, 2, 3, 5
Virahāṅka sequence

Discover the patterns

We will explore **height codes, even/odd parity, grid sums, magic squares, the Virahāṅka sequence, and letter–digit puzzles**. Every answer includes a method, not just the result.

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NCERT textbook questions and worked answers

How to use these pages

Cover the answer, try the question, then compare your steps. Page numbers refer to NCERT *Ganita Prakash*, Grade 7, Part I, Chapter 6. For diagram-based questions the solution is described with a reproducible arrangement or a fully specified grid.

Figure it Out I: heights in a queue (p. 128)

Q1 • Make six height-order patterns (a)–(f)

Answer. One set of valid arrangements is below. Each sequence lists seven **distinct heights**, left to right, using 1 for shortest and 7 for tallest. For each child count **earlier children who are taller**.

Required number calls	One possible height arrangement
(a) $0, 1, 1, 2, 4, 1, 5$	7, 3, 5, 4, 1, 6, 2
(b) $0, 0, 0, 0, 0, 0, 0$	1, 2, 3, 4, 5, 6, 7
(c) $0, 1, 2, 3, 4, 5, 6$	7, 6, 5, 4, 3, 2, 1
(d) $0, 1, 0, 1, 0, 1, 0$	2, 1, 4, 3, 6, 5, 7
(e) $0, 1, 1, 1, 1, 1, 1$	7, 1, 2, 3, 4, 5, 6
(f) $0, 0, 0, 3, 3, 3, 3$	5, 6, 7, 1, 2, 3, 4

Reasoning. Example (a), fifth child has height 1: the four before them are all taller, so that child says 4. Other valid height arrangements are possible.

Q2 • Always, sometimes or never? (a)–(f)

(a) **Sometimes.** Saying 0 only means nobody *in front* is taller.
(b) **Always.** The tallest child has no taller child in front.
(c) **Always.** The first child has nobody in front.
(d) **Sometimes.** A child in the middle can still have 0.
(e) **Sometimes.** The largest count need not belong to the shortest child; position matters.
(f) **7.** The eighth child can have seven taller children in front.

Figure it Out II: even and odd (p. 131)

Q1 • Parity of four sums

Answer. **(a) even; (b) even; (c) even; (d) even.**
Reasoning. Every even number contributes parity 0; two odd numbers cancel to even; eight odd numbers also pair into four even sums.

Q2 • Lakpa's coins total Rs. 205?

Answer. **Impossible.** An odd number of Rs. 1 coins contributes odd rupees; an odd number of Rs. 5 coins also contributes odd rupees. Their sum is **even**; the even number of Rs. 10 coins adds an even amount. The total must be even, but 205 is odd.

Q3 • Complete the four subtraction rules

Answer. (d) even − even = **even**; (e) odd − odd = **even**; (f) even − odd = **odd**; (g) odd − even = **odd**.
Reasoning. Subtraction has the same even/odd pattern as addition.

Figure it Out III: magic squares (p. 136)

Q1 • How many 1–9 magic squares?

Answer. Exactly **eight arrangements** if rotations and reflections are counted separately, or **one essential pattern** if arrangements differing only by rotation/reflection are considered the same. One example:

$$\begin{bmatrix} 8 & 1 & 6 \\ 3 & 5 & 7 \\ 4 & 9 & 2 \end{bmatrix} \quad \text{magic sum} = 15.$$

Q2 • Create a magic square with 2–10

Answer. Add 1 to *each cell* of the 1–9 square:

$$\begin{bmatrix} 9 & 2 & 7 \\ 4 & 6 & 8 \\ 5 & 10 & 3 \end{bmatrix} \quad \text{magic sum} = 18.$$

Reasoning. Each row has three cells, so increasing every cell by 1 increases every sum by 3.

Q3 • Increase every cell by 1 or double it

Answer. **(a)** Magic square remains; new magic sum $15 + 3 = 18$. **(b)** Magic square remains; new magic sum $2 \times 15 = 30$.
Reasoning. Each row, column and diagonal receives the same operation, so all eight sums still agree.

Q4 • Other operations giving magic squares

Answer. Add or subtract the **same constant** from every cell, or multiply/divide every cell by the **same nonzero constant**. Rotating or reflecting a magic square also preserves its sums. Applying different operations to different cells need not work.

Q5 • Any nine consecutive integers

Answer. Starting from $k, k+1, \ldots, k+8$, add $k-1$ to each entry of the 1–9 magic square. The new centre is $k+4$ and the magic sum is $3(k+4)$. For 9–17, add 8 to every cell and get sum 39.

Figure it Out IV: generalised magic squares (p. 137)

First derive the general pattern

If the middle number is m, the square can be written

$$\begin{bmatrix} m+3 & m-4 & m+1 \\ m-2 & m & m+2 \\ m-1 & m+4 & m-3 \end{bmatrix}.$$

For example, at $m = 5$ this is the familiar 1–9 magic square. Every row, column and diagonal sums to $3m$.

Q1 • Magic square whose centre is 25

Answer. Put $m = 25$ in the formula:

$$\begin{bmatrix} 28 & 21 & 26 \\ 23 & 25 & 27 \\ 24 & 29 & 22 \end{bmatrix}, \qquad \text{sum} = 75.$$

Q2 • Sum of any row, column or diagonal

Answer. $3m$. For instance, $(m+3)+(m-4)+(m+1) = 3m$.

Q3 • Add 1 to every cell; double every cell

Answer. **(a)** Centre becomes $m+1$, magic sum becomes $3m+3$.
(b) Centre becomes $2m$, magic sum becomes $6m$.
Reasoning. Apply the operation separately to each of the nine expressions, not to only the centre.

Q4 • Make a magic square of sum 60

Answer. $3m = 60$, so $m = 20$:

$$\begin{bmatrix} 23 & 16 & 21 \\ 18 & 20 & 22 \\ 19 & 24 & 17 \end{bmatrix}.$$

Q5 • Can the entries be non-consecutive?

Answer. **Yes.** Multiply every entry of the 1–9 magic square by 3:

$$\begin{bmatrix} 24 & 3 & 18 \\ 9 & 15 & 21 \\ 12 & 27 & 6 \end{bmatrix}, \qquad \text{sum} = 45.$$

Figure it Out V: final challenge (pp. 143–144)

Q1 • ON bulb switched 77 times

Answer. **OFF.** One toggle changes ON to OFF, the next changes OFF to ON. An *odd* number of toggles reverses the starting position.

Q2 • Can 50 two-sided loose sheets sum to 6000?

Answer. **No, assuming ordinary consecutively numbered front/back book pages.** Each sheet contains an odd page $2k-1$ and the following even page $2k$, so that sheet contributes $4k-1$, which leaves remainder 3 upon division by 4. For 50 sheets the total leaves remainder $50 \times 3 \equiv 2 \pmod 4$. But 6000 leaves remainder 0 (mod 4). Therefore it is impossible.
Important correction: The supplementary answer pages included with the supplied NCERT chapter say "possible" after checking only evenness. Checking remainder upon division by 4 proves that this conclusion is insufficient.

Q3 • Fill the 2×3 parity grid with three odds

Answer. One valid arrangement is:

O	E	E	**O**
O	E	O	**E**
E	**E**	**O**	

Reasoning. Row parities are odd, even; column parities are even, even, odd. Exactly three cells contain O.

Q4 • A magic square whose sum is zero

Answer. Subtract 5 from every cell of the 1–9 magic square:

$$\begin{bmatrix} 3 & -4 & 1 \\ -2 & 0 & 2 \\ -1 & 4 & -3 \end{bmatrix}, \qquad \text{sum} = 0.$$

All entries are not zero, as requested.

Q5 • Four even/odd blanks

Answer. (a) even; (b) even; (c) even; (d) odd. The number of odd terms determines whether a sum is odd; even terms do not alter its parity.

Q6 • Parity of $1 + 2 + \cdots + 100$

Answer. $\frac{100 \times 101}{2} = 5050$ is **even**. Alternative: there are fifty odd numbers from 1 to 100, an even number of odd terms.

Q7 • Around the terms 987, 1597

Answer. Next: $987 + 1597 = 2584$, then $1597 + 2584 = 4181$. Before: $1597 - 987 = 610$, then $987 - 610 = 377$. Thus

$$\boxed{377, 610, 987, 1597, 2584, 4181}.$$

Q8 • Ways to climb eight steps using 1s and 2s

Answer. 34 **ways.** Counting by the number of two-step jumps:

Two-step jumps	0	1	2	3	4
Ways	1	7	15	10	1

Reasoning. Sum $1 + 7 + 15 + 10 + 1 = 34$, the 8th number in the Virahāṅka sequence.

Q9 • Is the 20th Virahāṅka number odd or even?

Answer. Even. The parity sequence repeats **odd, even, odd** every three terms. Position $20 = 3 \times 6 + 2$ has the same parity as position 2: even.

Q10 • Which algebraic claims are true?

Answer. Only (a) is true, under the usual textbook use of positive counting-number letter values.

- **(a) True:** $4m - 1$ is odd, since $4m$ is even.
- **(b) False:** $6j - 4$ misses even numbers such as 4 for integer j.
- **(c) False for positive p, q:** $2p + 1$ misses 1, while $2q - 1$ includes it. With $p = 0$ allowed, both could represent all positive odds.
- **(d) False:** $2f + 3$ is always odd.

Q11 • Solve the cryptarithm $UT + TA = TAT$

Answer. $\boxed{U = 9,\ T = 1,\ A = 0}$. Substitution gives

$$\begin{array}{r} 91 \\ +10 \\ \hline 101 \end{array}$$

Reasoning. The three-digit answer begins with $T = 1$; the units sum is $T + A = T$, so $A = 0$. The tens column requires a carry to the hundreds place; hence $U = 9$.

Important questions inside the chapter (with answers)

p. 128 • The seven children in the picture

Answer. The textbook's given arrangement has counts $\boxed{0, 0, 1, 0, 3, 0, 3}$, read from left to right. In your own drawing, for each child count only taller children positioned before that child.

pp. 129–132 • More parity reasoning

Answer. Five odd-number cards **cannot** sum to 30 (five odd terms give an odd sum). Two siblings exactly one year apart cannot have ages summing to 112 (consecutive numbers sum to odd). Four odd terms sum to even; five to odd; six to even.
Small grid squares: 27×13 is odd, 42×78 is even, 135×654 is even. The product is odd only when both factors are odd.
Always-even examples: $2n$, $6n + 2$. **Always-odd:** $2n - 1$, $8n + 3$. **Either parity:** $3n + 4$. The 100th even number is 200 and the 100th odd number is 199.

p. 133 • Solve both row/column-sum grids

Answer. Here is one solution for each textbook grid (left, right):

$$\begin{bmatrix} 9 & 1 & 3 \\ 8 & 2 & 4 \\ 7 & 6 & 5 \end{bmatrix} \qquad \begin{bmatrix} 7 & 8 & 9 \\ 4 & 6 & 5 \\ 1 & 2 & 3 \end{bmatrix}$$

Left: rows $13, 14, 18$; columns $24, 9, 12$. **Right:** rows $24, 15, 6$; columns $12, 16, 17$. All digits from 1 to 9 occur once. The impossible grid on p. 134 includes a row sum 5 and column sum 26: any three distinct digits from 1–9 have a sum from 6 to 24.

pp. 137–138 • The Khajuraho square

Answer. The 4×4 *Chautīsā Yantra* has row, column and diagonal sum **34**. That explains its name. Other four-cell sets, such as its four corner values $7, 14, 9, 4$, also sum to 34.

pp. 140–142 • Rhythms and the Virahāṅka sequence

Answer. The number of rhythms for 1 to 8 beats is $1, 2, 3, 5, 8, 13, 21, 34$. After 89 come $144, 233, 377, 610$. Their parities repeat *odd, even, odd.* The supplementary answer transcript contains the typo "3779"; $144 + 233 = 377$ is correct.

p. 143 • Four extra letter-digit challenges

Answer. First, the textbook demonstrations: $T+T+T=UT$ gives $5+5+5=15$; $K2+K2=HMM$ gives $72+72=144$. The other four examples are below. Distinct letters represent distinct digits within each puzzle:

Puzzle	Filled-in sum
$YY+Z=ZOO$	$99+1=100$
$B5+3D=ED5$	$75+30=105$
$KP+KP=PRR$	$61+61=122$
$C1+C=1FF$	$91+9=100$

Learn the rules visually

1. Odd and even: picture the pairs



Even: all dots can pair
Odd: one dot remains

Parity rules you must remember

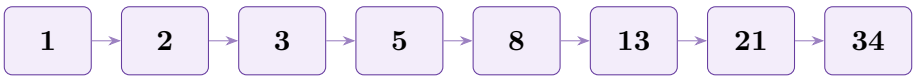
Addition/subtraction	Multiplication
$E \pm E = E$	$E \times$ anything $= E$
$O \pm O = E$	$O \times O = O$
$E \pm O = O$	**Odd product needs all odd factors.**

2. Magic square in colour

8	**1**	**6**
3	**5**	**7**
4	**9**	**2**

Every row, column and diagonal adds to 15

3. Why the Virahāṅka numbers grow



Next = previous two added

Common mistake

Magic square: equal rows alone are not enough: check columns *and both diagonals*. **Parity:** odd + odd is even, not odd. **Cryptarithms:** the same letter always denotes the same digit within one puzzle, and a leading digit cannot be zero.

Rapid revision, exam practice and answers

Five-minute self-test

1. Is 56×137 odd or even?
2. What is the 42nd odd number?
3. A 3×3 magic square has centre 11. What is its magic sum?
4. How many ways are there to make five beats with 1-beat and 2-beat syllables?
5. Is the 77th toggle of an initially ON light ON or OFF?

Answers: 1. Even. 2. $2(42) - 1 = 83$. 3. $3(11) = 33$. 4. 8 ways. 5. OFF.

Your memory map

HEIGHTS → count earlier taller children.
PARITY → only odd terms matter for a sum; one even factor makes a product even.
GRID → row-sum total equals column-sum total.
MAGIC → all row, column and diagonal sums equal 3× centre for 3-by-3 consecutive-number squares.
VIRAHĀṄKA → each term is the sum of the previous two.
CRYPTARITHM → solve columns from right to left and track carries.

Practice zone: questions worth revising

Try these in your notebook without looking at the answer key.

1. If n is odd, is $7n + 4$ odd or even?
2. Can seven odd numbers add up to 100? Explain.
3. A 3×3 consecutive-number magic square has middle entry 14. Find the sum of each row.
4. Continue: $13, 21, 34, 55, \ldots$ (three more terms).
5. A light is initially OFF. What happens after 120 toggles?
6. Which term describes $1, 3, 5, 7, \ldots$ at position n?

Answer key: (1) Odd: odd times odd plus even. (2) No: seven odds sum to odd. (3) 42. (4) $89, 144, 233$. (5) Still OFF: an even number of toggles. (6) $2n - 1$.